The State Is Unique: Quasi-Free Rigidity, the Closed Zero-Mode Door, and the Infrared Condition's Third Duty

Harald Rønning

Abstract. The squeeze-thermal state paper left its scope item (ii) open: whether ω_β is unique among $SO(3)$-invariant KMS states for the squeeze flow. We close it, in outline, with a stronger statement: ω_β is the unique regular β-KMS state of the squeeze flow on the infrared-smeared chiral algebra *outright* — the rotation-invariance constraint is not needed, because it is *derived*: the KMS condition forces the two-point operator to be a function of the one-particle generator, which commutes with rotations. Within the full quasi-free class (anomalous pairs and channel mixing included), uniqueness is proved directly: the KMS boundary condition yields two operator equations whose coefficients are strictly positive on the spectrum, killing anomalous correlations and pinning the gauge-invariant part to the Planck operator — precisely the covariance the construction used. Beyond the quasi-free class, the classical classification of KMS states for free Bose dynamics (Rocca–Sirugue–Testard) locates every non-uniqueness mechanism at the zero mode: condensate and chemical-potential families. That door is shut twice over here — spectrally, since the one-particle generator has trivial kernel by the Mellin conjugacy, and algebraically, since the infrared-smeared domain removes the zero mode from the algebra entirely. The computation makes the coincidence exact: the uniqueness equation degenerates *only* at $p = 0$, the very point the infrared condition excises. That condition thus performs its third structural duty — finiteness, necessity, and now uniqueness — the opposite profile of an ad hoc regulator. The result interlocks with the centralizer theorem: ω_β is an extreme point of the KMS simplex, and the simplex is the single point $\{\omega_\beta\}$.

---

1. The question, stated precisely

Fix the construction of the companion paper (*The Squeeze-Thermal State*): the infrared-smeared chiral Weyl algebra over the scale line, per angular channel — one-particle space with generator h = multiplication by p on $(0, \infty)$, absolutely continuous with trivial kernel by the Mellin conjugacy (Proposition 1 there) — with dynamics the translation flow, which *is* the squeeze flow in Mellin coordinates. Uniqueness is asked per pair (flow, β): the family over β is a reparametrization of the flow, not a rival state, since KMS at β for σ_t is KMS at 1 for $\sigma_{\beta t}$. The question: at fixed β, is ω_β the only KMS state?

2. Uniqueness within the full quasi-free class

A general quasi-free state is specified by a gauge-invariant two-point operator $\rho \geq 0$ and an anomalous (pair-correlation) part A; channel mixing and squeezing are permitted. Applying the KMS boundary condition to the two-point functions yields, in the spectral representation of h [standard; Manuceau–Verbeure lineage, Bratteli–

Robinson II], two operator equations.

Proposition 1 (no anomalous part). The anomalous equation reads $\big(1 - e^{-\beta(p+q)}\big) A(p,q) = 0$. The coefficient is strictly positive for all $p, q > 0$ (verified on the grid: minimum 2.0×10^{-4} at the grid edge, positive throughout, vanishing only in the limit $p+q \to 0$). Hence $A \equiv 0$: no squeezed KMS states exist.

Proposition 2 (the gauge-invariant part is pinned). The invariant equation reads $\rho\,(e^{\beta h} - 1) = 1$. Since $h > 0$ with trivial kernel, $e^{\beta h} - 1$ is invertible, and the solution is unique: $\rho = (e^{\beta h} - 1)^{-1}$, the Planck operator — equivalently the covariance $S(p) = \rho + \frac{1}{2} = \frac{1}{2}\coth(\beta p/2)$ of the companion construction, recovered to expected floating precision (4.3×10^{-9}, dominated by cancellation at small p).

Corollary (rotation-invariance is derived). The unique solution is a *function of h*: channel-mixing freedom among the degenerate angular channels is excluded automatically, and since h commutes with the $SO(3)$ action, the state is rotation-invariant *as a consequence of thermality*. Constraint (ii) of the original state problem was never independent.

The collapse is quantitative. Translation-invariance alone admits an infinite family: a deformed covariance $S + g$ with $g \geq 0$ remains a valid invariant quasi-free state (verified: positivity bound holds). Its KMS residual, however, is order one — computed maximum 2.6 — thermality, not symmetry, is what pins the state.

3. Beyond quasi-free: the zero-mode door, shut twice

For free Bose dynamics, the classical classification of KMS states (Rocca–Sirugue–Testard; the condensation literature descending from Araki–Woods; Bratteli–Robinson II) locates every mechanism of non-uniqueness at the zero mode of the one-particle generator: extremal KMS states beyond the quasi-free one are condensate-displaced or chemical-potential-shifted states — gauge-symmetry-breaking families living on ker-adjacent structure at $h = 0$.

Proposition 3 (the door is shut twice). *(i) Spectrally:* a condensate displacement requires a dynamics-invariant one-particle vector, an element of $\ker h$; by the Mellin conjugacy $\ker h = \{0\}$ — no zero eigenvalue exists. *(ii) Algebraically:* the infrared-smeared domain ($\hat{f}(0) = 0$) removes the zero mode from the algebra outright; the would-be shift automorphisms $W(f) \mapsto e^{iq\hat{f}(0)} W(f)$ act trivially, and there is no zero-mode observable for a condensate to displace. ■

The computation exhibits the two propositions as one fact: the coefficient $e^{\beta p} - 1$ of the uniqueness equation vanishes *only* at $p = 0$ — the sole point at which non-uniqueness could enter is exactly the point the infrared condition excises.

4. The infrared condition's third duty

The condition $\hat{f}(0) = 0$ now performs three structural jobs, established in three separate papers: it makes the thermal correlations *finite* (the construction paper), it is *necessary* — without it the two-point function diverges logarithmically, with increments matching β^{-1} ln 100 to four decimals (the centralizer paper, §6) — and it delivers *uniqueness* (here, Proposition 3(ii)). One scale-free condition, three duties: the precise opposite of an ad hoc regulator, and a further entry under the no-regulators remark of the construction paper's scope.

5. Conclusion, interlocked

Theorem (in outline). ω_β is the unique regular β-KMS state of the squeeze flow on the infrared-smeared chiral algebra; its rotation-invariance is automatic. (The regularity qualifier — the fields exist in the GNS representation — is the standard scoping of KMS classification for Weyl algebras, made explicit in the infrared write-out companion.) *Argument:* Propositions 1–2 give uniqueness within the full quasi-free class; the cited classification confines any further extremal KMS state to the zero mode; Proposition 3 shuts the zero mode twice over. The companion centralizer theorem supplies the cross-check: $M_{\omega_\beta} = \mathbb{C}1$ makes ω_β a factor state, hence an extreme point of the KMS simplex — and the above shows there are no others. The simplex is the single point $\{\omega_\beta\}$.

The state problem of this series thus ends with one state, not a choice among states: the algebra came from the contraction tower, the flow from the dilation's $\hbar$-preserving shadow, and thermality alone — given the forced spectrum — determines everything else, rotational isotropy included.

6. Scope

Two residuals, both deferred deliberately. *(i)* The functional-analytic setting, and the match to the classifications' stated hypotheses, are carried out in the companion note *The Infrared Write-Out*; the internal-lemma-level correspondence with the full text of Rocca–Sirugue–Testard remains a literature task, and until it is done the beyond-quasi-free part of the theorem inherits the cited classification's scope. *(ii)* Reconciliation with the thermal-state classification for conformal nets (Longo and collaborators): on nets that *retain* charge and zero-mode structure, chemical-potential families are the expected form of non-uniqueness, and their absence here should follow from the same domain condition — this is flagged as expected-and-to-be-verified, not asserted. Nothing in this note bears on the choice of β, which remains the one genuine physical parameter of the state.

References

- Companion papers: *The Squeeze-Thermal State: A Type-I No-Go, the Scale-Line Construction, and the Type III₁ Verdict*; *The Centralizer Is Trivial: Closing the Last Construction-Specific Link in the Type III₁ Verdict*; *The Second Contraction: From Algebraic R-Flux to the Heisenberg Algebra, the Dilation Symmetry It Creates, and the SO(3) Remnant of G2*; *The Infrared Write-Out: Test Space, Weyl Algebra, and the Verified KMS Structure on the Scale Line.*
- Rocca, F., Sirugue, M. & Testard, D., "On a class of equilibrium states under the Kubo–Martin–Schwinger condition. II. Bosons," *Comm. Math. Phys.* 19 (1970), 119–

141.

- Manuceau, J. & Verbeure, A., "Quasi-free states of the C.C.R.-algebra and Bogoliubov transformations," *Comm. Math. Phys.* 9 (1968), 293–302.
- Araki, H. & Woods, E. J., "Representations of the canonical commutation relations describing a nonrelativistic infinite free Bose gas," *J. Math. Phys.* 4 (1963), 637–662.
- Lewis, J. T. & Pulé, J. V., "The equilibrium states of the free boson gas," *Comm. Math. Phys.* 36 (1974), 1–18.
- Bratteli, O. & Robinson, D. W., *Operator Algebras and Quantum Statistical Mechanics II*, Springer (1981; 2nd ed. 1996).

Created in collaboration with Claude (Anthropic). All conclusions and any errors remain the author's responsibility.